

## PROBLEM SET 2—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

**Instructions:** Due on 22/10/2020; scan or type your work and submit through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

1. A monopolist can produce a good in different qualities. The cost of producing a unit of quality  $s$  is  $s^3$ . Consumers buy at most one unit and have valuation function:

$$v(s, \theta) = \begin{cases} \theta s & \text{if they consume one unit of quality } s, \\ 0 & \text{if they do not consume.} \end{cases}$$

The monopolist decides on the quality (or qualities) it is going to produce and price. Consumers observe qualities and prices and decide which quality to buy if at all. Denote the price (or prices) that the monopolist charges the consumer by  $P$ .

- (a) Interpret the parameter  $\theta$  of the consumers.
- (b) Characterize the first-best solution, assuming that  $\theta$  belongs to a subset  $\Theta \subset \mathbb{R}$ .
- (c) How do the first-best quality depends on  $\theta$ ?
- (d) Now assume that  $\Theta = \{\theta_L, \theta_H\}$  with  $\theta_L < \theta_H$ . Furthermore, suppose that the seller cannot observe  $\theta$  but believes that  $\theta = \theta_H$  with probability  $\beta$  and that  $\theta = \theta_L$  with probability  $1 - \beta$ . Write the optimization program that the monopolist will need to solve.
- (e) Characterize the second-best solution and the consumer's optimal informational rents.

### Solution

- (a) Assume that  $\theta \in \mathbb{R}_+$ . Notice that  $\frac{\partial v(s|\theta)}{\partial s} = \theta$  so that the marginal utility with respect to quality depends on the parameter  $\theta$ . Thus, the larger the consumer's  $\theta$ -parameter, the larger it is the marginal utility with respect to quality.

- (b) For the First-Best, recall that for each type, the principal solves

$$\begin{aligned} \max_{s_\theta, P_\theta} \{ & P_\theta - s_\theta^3 \} \\ \text{s.t. } & \theta s_\theta - P_\theta \geq 0 \end{aligned} \tag{1}$$

where  $(s_\theta, P_\theta)$  is a type-dependent menu of qualities and transfers. Note that the participation constraint will bind, i.e.  $P_\theta = \theta s_\theta$  and thus, the principal can solve the program:

$$\max_{q(\theta)} \{ \theta s_\theta - s_\theta^3 \} \tag{2}$$

The first order condition for this program is

$$\theta = 3(s_\theta^f)^2 \Rightarrow s_\theta^f = \sqrt{\frac{\theta}{3}},$$

which characterizes the optimal quantity. The previous expression together with  $P_\theta^f = \theta s_\theta^f$  characterize the optimal first-best contract.

- (c) For the First-Best quality

$$\frac{ds_\theta^f}{d\theta} = \frac{1}{2\sqrt{3\theta}} > 0$$

for every  $\theta \in \mathbb{R}_+$ . Thus quality is increasing with  $\theta$ .

(d) The monopolist needs to solve the following program:

$$\begin{aligned}
& \max_{\{(s_L, s_H), (P_L, P_H)\}} \{\beta(P_H - s_H^3) + (1 - \beta)(P_L - s_L^3)\} \\
& \text{s.t.} \quad \theta_H s_H - P_H \geq \theta_H s_L - P_L \\
& \quad \theta_L s_L - P_L \geq \theta_L s_H - P_H \\
& \quad \theta_H s_H - P_H \geq 0 \\
& \quad \theta_L s_L - P_L \geq 0.
\end{aligned} \tag{3}$$

(e) In problem (3), the participation constraint for the low type and the incentive compatibility constraint for the high type imply the participation constraint for the high type. Furthermore, the ability of the high type to mimic the low type but not the other way around implies that the incentive compatibility constraint of the low type will be satisfied and thus can be omitted. Indeed, consider the first-best contract, which involves zero rents for both types, that contract is not incentive compatible because the  $\theta_H$  buyer will prefer to choose the contract for the low type than her own first-best allocation: while this allocation restricts consumption, it nevertheless allows the high type to experience a positive informational rent. On the other hand, the low type  $\theta_L$  will not find attractive to raise his consumption to the first best, since doing so would involve a transfer that, while leaving the high type with zero utility, would lead to a low type to experience a negative payoff since he has lower valuation for consumption.

Furthermore, the participation constraints for the low type and the incentive constraint for the high type will hold with equality:

$$\theta_H s_H - P_H = \theta_H s_L - P_L, \quad P_L = \theta_L s_L.$$

Solving for transfers and substituting them into problem (3) yields

$$\max_{(q_H, q_L)} \beta(\theta_H s_H - (\theta_H - \theta_L)s_L - s_H^3) + (1 - \beta)(\theta_L s_L - s_L^3). \tag{4}$$

The first order conditions for program (4) yield:

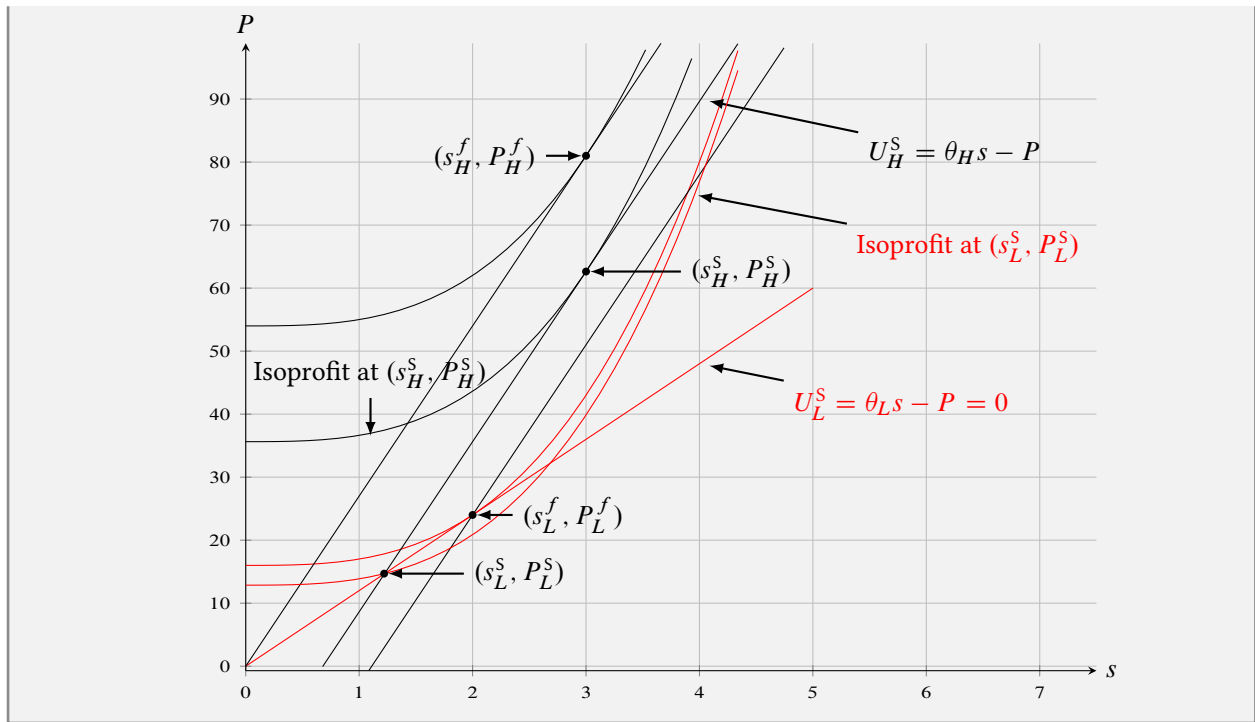
$$\begin{aligned}
& \theta_H - 3(s_H^S)^2 = 0 \Rightarrow s_H^S = \sqrt{\frac{\theta_H}{3}} \\
& -\beta(\theta_H - \theta_L) + (1 - \beta)(\theta_L - 3(s_L^S)^2) \leq 0 \Rightarrow s_L^S = \max \left\{ \sqrt{\frac{\theta_L}{3} - \frac{\beta}{3(1 - \beta)}(\theta_H - \theta_L)}, 0 \right\},
\end{aligned}$$

where the last equation accounts for the fact that  $s_L^S$  can have a corner solution, i.e., a binding non-negativity constraint. Prices are given by

$$\begin{aligned}
P_L^S &= \theta_L \max\{s_L^S, 0\}, \\
P_H^S &= \theta_H s_H^S - (\theta_H - \theta_L)s_L^S.
\end{aligned}$$

The informational rent for the low type agent is zero and for the high type agent is  $(\theta_H - \theta_L)s_L^S \geq 0$  with strict inequality as long as  $s_L^S$  is strictly positive.

Here is a graph of the solution when  $\theta_H = 27$ ,  $\theta_L = 12$  and  $\beta = 1/3$ . With these parameter values, we have a strictly positive solution for  $s_L^S$ .



2. Consider the model we saw in lecture 7 about credit rationing. For completeness sake, I write here again the main description. Consider a population of risk-neutral borrowers who each own a project that requires an initial outlay of  $I = 1$  and yields a random return  $X$ , where  $X \in \{0, R\}$ . Let  $p \in [0, 1]$  denote the probability that  $X = R$ . Borrowers have no wealth and must obtain investment funds from a bank. A borrower can be of two different types  $i = s, r$ . The borrower of type  $i$  has project with return characteristics  $(p_i, R_i)$ . Assume the following.

A1'  $p_r R_r < 1 < p_s R_s$ .

A2  $p_s > p_r$  and  $R_s < R_r$ .

As you can see, assumption A1' is different from the one we covered in class. Assume that there is excess demand for funds for the bank, i.e.  $\alpha < 1$  and that available funds are sufficient to avoid crowding out either type of borrower completely, i.e.  $\max\{\beta, 1 - \beta\} < \alpha$ .

- Compute the optimal single fixed repayment scheme  $D$  for the bank. Is there any type of borrower that is excluded for sure?
- Now consider that the bank can offer a type-dependent contract that specifies the probability of financing,  $x_i$  and the repayment amount  $D_i$  if the project is successful. That is, the bank can offer a plan  $(x_i, D_i)$  for  $i = s, r$ . Compute the optimal contracts for the bank under assumptions A1' and A2. Show that in contrast with what we saw in class, now credit rationing reappears under two possible forms:
  - If  $R_s[\beta p_s + (1 - \beta)p_r] \geq 1$  then both types of project are funded and there is cross-subsidization between types. Conclude that this happens because the banks cannot turn away risky borrowers without restricting access to credit for the safe borrowers.
  - If  $R_s[\beta p_s + (1 - \beta)p_r] < 1$  then neither project is funded, which is a severe form of credit rationing, i.e. *total financial collapse*.

#### Solution

- Let  $D$  be a single-fixed repayment scheme. As seen in the lecture notes, type- $i$  borrowers apply for funds if and only if  $D \leq R_i$ . If  $D > R_s$  only the risky borrowers will consider applying. The maximum then that the bank can charge is  $D = R_r$ . In this case, the bank is only contracting with risky borrowers and obtains  $(1 - \beta)(p_r R_r - 1)$ . In contrast with the case analyzed in class, assumption A1' implies that this term is negative. Thus, contracting with risky borrowers only is not optimal. This implies that

$D \leq R_s$ , i.e. the bank should contract with both types, assuming of course that this is profitable. If this is the case, the bank should charge the highest possible repayment that the safe borrowers can afford, that is  $D = R_s$ . In this case, the profits obtained by the bank are given by

$$\alpha(\beta(p_s R_s - 1) + (1 - \beta)(p_r R_s - 1)).$$

In order for the bank to be willing to contract with both borrowers, the expression above must be positive. Thus, profits are positive if and only if  $R_s(\beta p_s + (1 - \beta)p_r) \geq 1$  and in this case the bank finds optimal to contract with both types. On the other hand, if  $R_s(\beta p_s + (1 - \beta)p_r) < 1$  the bank abstains from lending to any of the borrowers. Note that in both situations, there is credit rationing.

- (b) We need to characterize the second-best contract for the bank. A contract is defined by  $(x_i, D_i)$  where  $x_i$  is the probability of obtaining financing and  $D_i$  is the amount to be repaid. The bank's problem is:

$$\max_{(x_i, D_i)_{i=s,r}} \{ \beta x_s (p_s D_s - 1) + (1 - \beta) x_r (p_r D_r - 1) \} \quad (5)$$

$$\text{subject to } 0 \leq x_i \leq 1, \quad \forall i = s, r \quad (6)$$

$$\beta x_s + (1 - \beta) x_r \leq \alpha \quad (7)$$

$$x_i p_i (R_i - D_i) \geq 0 \quad \forall i = s, r \quad (8)$$

$$x_i p_i (R_i - D_i) \geq x_j p_i (R_i - D_j) \quad \forall i, j = s, r \quad (9)$$

Because of assumption A2, the risky type still has the highest willingness to repay, so it is still considered the "high type". Therefore, the previous problem is equivalent to the reduced problem:

$$\max_{(x_i, D_i)_{i=s,r}} \{ \beta x_s (p_s D_s - 1) + (1 - \beta) x_r (p_r D_r - 1) \} \quad (10)$$

$$\text{subject to } x_s p_s (R_s - D_s) \geq 0 \quad (11)$$

$$x_r p_r (R_r - D_r) \geq x_s p_r (R_r - D_s) \quad (12)$$

$$x_r \geq x_s \quad (13)$$

$$0 \leq x_i \leq 1 \quad i = s, r \quad (14)$$

$$\beta x_s + (1 - \beta) x_r \leq \alpha \quad (15)$$

Furthermore, we know that the participation constraint for the safe type and the incentive constraint for the risky type will be binding. Solving for  $D_s$  and  $D_r$  using these binding constraints and plugging the resultant expressions into the objective yields the following problem:

$$\max_{x_s, x_r} x_s (\beta(p_s R_s - 1) - (1 - \beta)p_r(R_r - R_s)) + x_r ((1 - \beta)(p_r R_r - 1)) \quad (16)$$

$$\text{subject to } 0 \leq x_s \leq x_r \leq 1 \quad (17)$$

$$\beta x_s + (1 - \beta) x_r \leq \alpha \quad (18)$$

Notice that in the objective (16), the term attached to  $x_r$  is strictly negative. Thus the bank would like to shutdown any interaction with the risky types. However, if  $x_r = 0$  then it would force  $x_s$  to be zero also because of the monotonicity constraint. Since it might still be profitable to contract with the safe type, then  $x_s = 0$  might not always be optimal. This implies that the monotonicity constraint will be binding. To see this, suppose towards a contradiction that  $\hat{x}_s$  and  $\hat{x}_r$  are optimal solutions such that  $\hat{x}_r = \hat{x}_s + \epsilon$  where  $\epsilon > 0$ . The expected profit then is

$$\hat{x}_s [R_s(\beta p_s + (1 - \beta)p_r) - 1] + \epsilon(1 - \beta)(p_r R_r - 1).$$

However, by taking instead  $\tilde{x}_r = \hat{x}_s$  we have that the profit from the contract  $(\hat{x}_s, \tilde{x}_r)$  is such that

$$\hat{x}_s [R_s(\beta p_s + (1 - \beta)p_r) - 1] + \epsilon(1 - \beta)(p_r R_r - 1) < \hat{x}_s [R_s(\beta p_s + (1 - \beta)p_r) - 1]$$

which contradicts the optimality of  $\hat{x}_r$ . Thus, the monotonicity constraint must be binding.

Let  $x = x_r = x_s$ . Then the problem reduces to

$$\max_{x \in [0, \alpha]} x [R_s(\beta p_s + (1 - \beta)p_r) - 1] \quad (19)$$

From the previous problem it is easy to see that we have the following optimal solution

$$x^* = \begin{cases} \alpha & \text{if } R_s(\beta p_s + (1 - \beta)p_r) \geq 1, \\ 0 & \text{if } R_s(\beta p_s + (1 - \beta)p_r) < 1. \end{cases}$$

With this solution, it is easy to find the optimal repayment scheme from the binding incentive and participation constraints:  $D_r = D_s = D^* = R_s$ . Thus, the optimal contract is a *pooling contract* and it is the same as the one found in the previous item. Furthermore, the solution implies that if the bank decides to actually fund any of the consumers, it is cross-subsidizing the risky types with the profits obtained from the safe ones. If this is the case, there is still credit rationing because the probability of funding any type of consumer is  $\alpha$ . When the bank decides not to contract with any type, there is a total financial collapse, since no consumer gets credit.

3. (J. Tirole) Assume that there are two types of consumers for a firm's product,  $\theta_H$  and  $\theta_L$ . The proportion of type  $\theta_L$  consumers is  $\lambda$ . A type  $\theta$ 's utility when consuming amount  $x$  of the good and paying a total  $T$  for it is  $u(x, T) = \theta v(x) - T$ , where

$$v(x) = \frac{1 - (1 - x)^2}{2}.$$

The firm is the sole producer of this good, and its cost of production per unit is  $c > 0$ .

- Consider a non-discriminating monopolist. Derive his optimal pricing policy, i.e. the cost per unit that the monopolist will charge. Show that he serves both classes of consumers if either  $\theta_L$  or  $\lambda$  is "large enough." *Hint: Derive the demand curve for each type as a function of the price charged by the monopolist and then compute the "aggregate demand". Use this demand to solve for the optimal price of the monopolist.*
- Consider a monopolist who can distinguish the two types (by some characteristic) but can only charge a simple price  $p_i$  for each type  $\theta_i$ . Characterize the optimal prices. Show that at the optimal prices, the monopolist is charging a marginal price  $p = c$  per unit plus a personalized fixed fee equal to the consumer's *net surplus*.
- Suppose that the monopolist cannot distinguish the types. Suppose that the monopolist can charge a single *two-part tariff*. A two part tariff is an *affine pricing schedule*:  $T(q) = A + pq$ , where  $A$  is a fixed fee and  $p$  is a marginal price per unit. Derive the optimal two part tariff under the assumption that the monopolist serves both types.
- Compute the fully optimal nonlinear tariff (second best optimum for the monopolist's screening problem). How do the quantities purchased by the two types compare with the levels obtained in (a)–(c). How do the monopolist's profits from each case in (a)–(d) compare?

#### Solution

It will be useful to calculate some expressions that will be used repeatedly. For each type, the first order condition in the utility maximization problem of the consumer when he faces a *single-posted price* yields the individual demands:

$$\theta_i(1 - x_i) = p_i \Rightarrow x_i(p) = \max \left\{ 1 - \frac{p}{\theta_i}, 0 \right\} \quad \text{for } i = L, H. \quad (20)$$

Assume that  $\theta_H > \theta_L > c > 0$  in order to simplify the analysis by eliminating the cases in which no consumer would like to buy at the lowest price the monopolist can produce without incurring in losses.

In order to simplify calculations, let  $\theta$  denote the *harmonic mean* of  $\theta_L$  and  $\theta_H$ :

$$\frac{1}{\theta} \equiv \frac{\lambda}{\theta_L} + \frac{1 - \lambda}{\theta_H}. \quad (21)$$

Then the aggregate demand at any price  $p$  can be written as

$$X(p) = \lambda x_L(p) + (1 - \lambda)x_H(p) = 1 - \frac{p}{\theta}. \quad (22)$$

Net consumer surplus at a price  $p$  is given by

$$S_i(p) = \theta_i v(x_i(p)) - p x_i(p)$$

so that for the specific functional form that we assumed for the utility the previous expression yields

$$S_i(p) = \theta_i \left( \frac{1 - (1 - x_i(p))^2}{2} \right) - p x_i(p) = \frac{(\theta_i - p)^2}{2\theta_i}. \quad (23)$$

Note that  $S_i \geq 0$  and that surplus is always higher for  $\theta_H$  types.

(a) The monopolist that can only charge a single price  $p$  would solve the following problem

$$\Pi_1 = \max_{p \geq 0} (p - c)X(p) \quad (24)$$

Since  $X(p) \leq 0$  for any  $p \geq \theta$  and the objective is strictly concave in  $p$ , the first order condition is necessary and sufficient to identify a solution to the monopolist's problem, which yields the monopoly price

$$p_1 = \frac{\theta - c}{2}$$

and the monopoly profits

$$\Pi_1 = \frac{(\theta - c)^2}{4\theta}.$$

Note that this solution assumes that the monopolist chooses to serve the two types of consumers. The monopolist could potentially choose a price high enough such type  $\theta_L$  consumers choose not to buy. Such a strategy is optimal if the monopoly price relative to this category, which is  $\frac{c + \theta_H}{2}$ , exceeds  $\theta_L$  and if the profits from this strategy are larger than the monopoly profits of serving the two types. Serving only the high type at monopoly price  $(c + \theta_H)/2$  yields

$$\Pi'_1 = (1 - \lambda) \frac{(\theta_H - c)^2}{4\theta_H}.$$

It can be shown that<sup>a</sup>  $\frac{\partial(\Pi_1 - \Pi'_1)}{\partial \lambda} > 0$ . Furthermore, it is easy to see that  $\lim_{\lambda \rightarrow 0} (\Pi_1 - \Pi'_1) < 0$  and that  $\lim_{\lambda \rightarrow 1} (\Pi_1 - \Pi'_1) > 0$  and thus, by monotonicity and the intermediate value theorem,  $(\Pi_1 - \Pi'_1)$  is equal to zero at just one point  $\lambda^*$ . Thus, for  $\lambda \geq \lambda^*$ , i.e. if  $\lambda$  is "high enough", serving the two types of consumers is more profitable than only serving the high types. Alternatively, if  $\theta_L > (c + \theta_H)/2$ , monopoly pricing for the type  $\theta_H$  consumers yields consumption for the type  $\theta_L$  consumers.

(b) If the monopolist can distinguish between the two types then we are calculating the first-best outcome. Thus, the pair of problems of the perfectly discriminating monopolist are given by

$$\max_{x_i, T_i} T_i - c x_i \quad \text{subject to } \theta_i v(x_i) - T_i \geq 0 \text{ for } i = L, H.$$

As usual, the participation constraint must be binding and we can simplify the pair of problems to the following ones:

$$\max_{x_i, T_i} \theta_i v(x_i) - c x_i \quad \text{subject to } \theta_i v(x_i) - T_i \geq 0 \text{ for } i = L, H.$$

Notice that the monopolist is maximizing social surplus for each type. Due to the specific form for  $v$  that we assumed, the objective is strictly concave and continuous and therefore, the first order conditions are necessary and sufficient. From them we get

$$\theta_i (1 - x_i^f) = c \Rightarrow x_i^f = 1 - \frac{c}{\theta_i}, i = L, H.$$

At this solution, we have that  $T_i^f = \theta_i v(x_i^f) = \frac{(\theta_i - c)(\theta_i + c)}{2\theta_i}$ . Notice that this transfer is equivalent to the monopolist charging a marginal price of  $p = c$  per unit and also charging a fixed fee equal to the net surplus of each consumer  $S_i(c) = \frac{(\theta_i - c)^2}{2\theta_i}$ . Notice that

$$S_i(c) + cx_i^f = \frac{(\theta_i - c)^2}{2\theta_i} + cx_i(p) = \frac{(\theta_i - c)^2}{2\theta_i} + \frac{\theta_i c - c^2}{\theta_i} = \frac{\theta_i^2 - c^2}{2\theta_i} = \frac{(\theta_i - c)(\theta_i + c)}{2\theta_i} = T_i^f.$$

Profits from the first best are

$$\Pi_2 = \lambda(T_L^f - cx_L^f) + (1 - \lambda)(T_H^f - cx_H^f) = \frac{\lambda(\theta_L - c)^2}{2\theta_L} + \frac{(1 - \lambda)(\theta_H - c)^2}{2\theta_H}.$$

- (c) Now we look for the optimal single two-part tariff. Again, for simplicity, we will assume that the monopolist serves the two types of consumers, i.e. that  $\lambda$  and  $\theta_L$  are sufficiently large, as we saw before.

Suppose that the monopolist charges a marginal price of  $p$ . The highest fixed fee that is consistent with type  $\theta_L$  consumers buying the good is  $A = S_L(p)$ . Type  $\theta_H$  consumers then buy because as mentioned before  $S_H(p) > S_L(p) = A$ . Thus, the monopolist's problem is

$$\max_{p \in [0, \infty)} S_L(p) + (p - c)X(p).$$

The monopolist will do at least as well under the optimal single two-part tariff as under a fully linear tariff (single monopoly price). To the variable profit  $(p - c)X(p)$  the monopolist is adding the fixed fee  $S_L(p)$  that is paid by all types of consumers. Alternatively, notice that a linear tariff is a special case of a two-part tariff.

The first order condition yields

$$\frac{p - \theta_L}{\theta_L} - \frac{p}{\theta} + \frac{c}{\theta} + 1 - \frac{p}{\theta} = 0.$$

Solving the previous equation yields  $p_3 = \frac{c}{2 - \frac{\theta}{\theta_L}}$ . The monopolist profit, after a lot of tedious algebra is

$$\Pi_3 = \frac{1}{4} \left( c^2 \left( \frac{1}{\theta} + \frac{1}{2\theta_L - \theta} \right) - 4c + 2\theta_L \right).$$

- (d) Finally, for the optimal non-linear tariff, we follow the standard procedure. From the general problem analyzed in class, we know that the monopolist will solve

$$\max_{(x_i, T_i)_{i=L, H}} \lambda(T_L - cx_L) + (1 - \lambda)(T_H - cx_H) \quad (25)$$

$$\text{subject to } \theta_H v(x_H) - T_H \geq \theta_H v(x_L) - T_L \quad (26)$$

$$\theta_L v(x_L) - T_L \geq \theta_L v(x_H) - T_H \quad (27)$$

$$\theta_H v(x_H) - T_H \geq 0 \quad (28)$$

$$\theta_L v(x_L) - T_L \geq 0 \quad (29)$$

$$x_L \geq 0, x_H \geq 0 \quad (30)$$

This problem is equivalent to the reduced problem in which we drop the incentive constraint of the low type, the participation constraint of the high type and we impose a monotonicity constraint. Furthermore, the incentive constraint of the high type and the participation constraint of the low type need to be binding. Using these facts and substituting the binding constraints into the objective and rewriting it we can solve the following problem:

$$\max_{x_L \geq 0, x_H \geq 0} \theta_L v(x_L) - cx_L + \frac{1 - \lambda}{\lambda} (\theta_H v(x_H) - cx_H) - \frac{1 - \lambda}{\lambda} (\theta_H v(x_L) - \theta_L v(x_L))$$

Notice that in the previous problem, as usual we have dropped the monotonicity constraint. We will check later that it holds.

From the first order conditions we get that

$$\theta_H(1 - x_H^s) = c \Rightarrow x_H^s = 1 - \frac{c}{\theta_H},$$

thus we have once again that there is no distortion at the top, and the second best quantity for the high type is the same as the first best.

For the low type:

$$\theta_L(1 - x_L^s) - c - \frac{1 - \lambda}{\lambda}(\theta_H(1 - x_L^s) - \theta_L(1 - x_L^s)) = 0$$

which implies that

$$x_L^s = 1 - \frac{c}{\theta_L - \frac{1 - \lambda}{\lambda} \Delta \theta}$$

where  $\Delta \theta = \theta_H - \theta_L$ . Notice that indeed  $x_L^s \leq x_H^s$  and thus the monotonicity constraint is satisfied. The optimal transfers are

$$T_L^s = \theta_L v(x_L^s), \quad T_H^s = \theta_H(v(x_H^s) - v(x_L^s)) + \theta_L v(x_L^s).$$

Let  $\Pi_4$  be the optimal non-linear tariff.

It is clear from the way we solved the problem, (more restricted to less restricted) that  $\Pi_1 \leq \Pi_3 \leq \Pi_4 \leq \Pi_2$ .

<sup>a</sup>  **Warning!** The following part requires a ridiculous amount of algebra. But for the sake of completeness I will write here the proof that the derivative with respect to  $\lambda$  of the profit difference  $\Pi_1 - \Pi_1'$  is strictly positive and thus increasing in  $\lambda$ . Calculating the derivative we get

$$\frac{\partial(\Pi_1 - \Pi_1')}{\partial \lambda} = \frac{(\theta_H - \theta_L) \left( \frac{\theta_H \theta_L}{\lambda \theta_H - \lambda \theta_L + \theta_L} - c \right)^2}{4 \theta_H \theta_L} + \frac{(\theta_H - \theta_L) \left( \frac{\theta_H \theta_L}{\lambda \theta_H - \lambda \theta_L + \theta_L} - c \right)}{2 (\lambda \theta_H - \lambda \theta_L + \theta_L)}.$$

Expanding all terms, simplifying and canceling terms, then factoring and collecting similar terms yields:

$$\frac{1}{4} \left( c \left( \frac{c}{\theta_L} - 2 \right) - \frac{\theta_H^2 \theta_L}{(\lambda \theta_H - (\lambda - 1) \theta_L)^2} + \theta_H \left( \frac{\theta_L^2}{(\lambda \theta_H - \lambda \theta_L + \theta_L)^2} + 1 \right) \right).$$

We want to show that the previous expression is greater than zero to show that the difference in profits is strictly increasing with respect to  $\lambda$ . The proof will now proceed in two steps: we will show that when  $\lambda \rightarrow 0$ , the term that we obtain is strictly positive. Next we will show that the derivative is also an increasing function with respect to  $\lambda$  by calculating the second derivative and showing that it is positive. These two facts will allow us to conclude that the first derivative is strictly increasing and thus always strictly positive over its domain.

First note that when  $\lambda \rightarrow 0$ , the previous expression evaluates to

$$\frac{(c - \theta_H)(c + \theta_H - 2\theta_L)}{\theta_L}.$$

The previous term is strictly positive if and only if  $c + \theta_H - 2\theta_L < 0$  which is exactly the condition that the monopoly price with respect to the type- $\theta_H$  market is less than the highest willingness to pay of type- $\theta_H$ . Now, calculating the second derivative with respect to  $\lambda$  yields:

$$\frac{\theta_H \theta_L (\theta_H - \theta_L)^2}{2 (\lambda \theta_H - \lambda \theta_L + \theta_L)^3},$$

which once again is easy to see is strictly positive. Hence, the first derivative is an increasing function and is strictly positive in all of its domain which concludes the proof.

4. Wuzzle Airlines is the only airline allowed to fly between the Isle of Wuzz and the country of Panda-La. There are two types of passengers, tourists and business. Business travelers are willing to pay more than tourists. The airline, however, cannot tell directly whether a ticket purchaser is a tourist or a business traveler. The two types differ, though, in how much they are willing to pay to avoid having to purchase their tickets in advance. (Passengers do not like to commit themselves in advance to traveling at a particular time.)

More specifically, the utility levels of each of the two types net of the price of the ticket,  $P$ , for any

given amount of time  $W$  prior to the flight that the ticket is purchased are given by

$$\begin{aligned}\text{Business: } & v - \theta_B P - W \\ \text{Tourist: } & v - \theta_T P - W,\end{aligned}$$

where  $0 < \theta_B < \theta_T$ . (Note that for any given level  $W$ , the business traveler is willing to pay more for his ticket. Also, the business traveler is willing to pay more for any given reduction in  $W$ .)

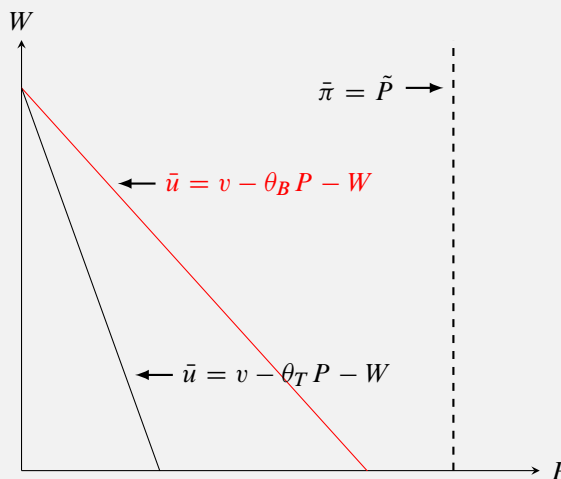
The proportion of travelers who are tourists is  $\lambda$ . Assume that the cost of transporting a passenger is  $c$ .

Assume in (a)–(d) that Wuzzle Airlines wants to carry both types of passengers.

- Draw the indifference curves of the two types in  $(P, W)$ -space. Draw the airline's isoprofit curves. Now formulate the optimal (profit-maximizing) price discrimination problem mathematically that Wuzzle Airlines would want to solve. *Hint: Impose non negativity of prices as a constraint since, if it charged a negative price, it would sell an infinite number of tickets at this price. Waiting times also need to be nonnegative.*
- Show that in the optimal solution, tourists are indifferent between buying a ticket and not going at all.
- Show that in the optimal solution, business travelers never buy their ticket prior to the flight and are just indifferent between doing this and buying when tourists buy.
- Describe fully the optimal price discrimination scheme under the assumption that they sell to both types. How does it depend on the underlying parameters  $\lambda$ ,  $\theta_B$ ,  $\theta_T$  and  $c$ ? Provide an economic interpretation.
- Under what circumstances will Wuzzle Airlines choose to serve only business travelers?

### Solution

- The following diagram plots the indifference curves of the two types of travelers and the iso-profit line of the airline in  $(P, W)$ -space.



The airline only pays cost  $c$  of a passenger if it chooses to provide service, i.e. if it chooses to charge at least one strictly positive price. If by charging at least one strictly positive price the airline finds that it cannot make a profit, then it decides not to operate. Thus, the problem of the airline is

$$\begin{aligned}
& \max_{\{(P_T, W_T), (P_B, W_B)\}} \max\{\lambda P_T + (1 - \lambda) P_B - c, 0\} & (31) \\
& \text{subject to } \theta_B P_B + W_B \leq \theta_B P_T + W_T & (\text{IC B}) \\
& \theta_T P_T + W_T \leq \theta_T P_B + W_B & (\text{IC T}) \\
& v \geq \theta_B P_B + W_B & (\text{PC B}) \\
& v \geq \theta_T P_T + W_T & (\text{PC T}) \\
& P_T \geq 0, P_B \geq 0, W_T \geq 0, W_B \geq 0 & (\text{Non-Negativity})
\end{aligned}$$

The airline's problem is written in that way because if it decides not to operate, then it doesn't need to pay a cost. The airline doesn't operate when it charges both prices equal to zero. In order to simplify this problem, we will assume for the moment that the airline decides to operate always, i.e. that profit is always positive. Thus we will consider as an objective

$$\lambda P_T + (1 - \lambda) P_B - c,$$

and we will worry later about in what circumstances the firm will make negative profits.

- (b) In the maximization problem above we have that (PC T) and (IC B) together with  $\theta_B < \theta_T$  (which in this problem is the single-crossing property) imply that constraint (PC B) is redundant, so it is never binding. constraint (PC T) must therefore be binding for if it would not, then we can reduce  $P_T$  and  $P_B$  by a small positive quantity  $\epsilon > 0$  and all the remaining constraints will still be satisfied and this would improve the airline's profit. The fact that (PC T) is binding implies that tourists are indifferent between buying and not buying a ticket.
- (c) Suppose that  $\{(P_T, W_T), (P_B, W_B)\}$  is an optimal, incentive compatible contract. We need to show that  $W_B = 0$ . Towards a contradiction, assume that  $W_B > 0$ . Choose  $\epsilon > 0$  to reduce  $W_B$  by  $\epsilon$  such that it still remains positive and increase  $P_B$  by  $\epsilon/\theta_B$  so that the utility of type B does not change. Now the firm earns higher profits from the business type. The actual change in profit is  $(1 - \lambda)\epsilon/\theta_B > 0$ . Now we need to check that this new contract we constructed is not attractive for the tourist type. Indeed this is the case:

$$\theta_T P_T + W_T \leq \theta_T P_B + W_B = \theta_T \left( P_B + \frac{\epsilon}{\theta_B} \right) + (W_B - \epsilon) < \theta_T \left( P_B + \frac{\epsilon}{\theta_B} \right) + (W_B - \epsilon),$$

where the first inequality comes from  $\{(P_T, W_T), (P_B, W_B)\}$  being an incentive compatible plan, the second equality comes from altering  $W_B$  by  $\epsilon$  but then increasing  $P_B$  by  $\epsilon/\theta_B$  so as to not alter the utility of the tourist type, and the third inequality comes from the fact that  $\theta_B < \theta_T$ . The previous calculations shows that the altered contract is still incentive compatible but it achieves a higher profit for the airline, contradicting the optimality of  $\{(P_T, W_T), (P_B, W_B)\}$  which we assumed originally. Thus we must have that  $W_B = 0$  at an optimal incentive compatible plan.

Now, if at an optimal contract, business travelers were not indifferent between  $(P_T, W_T)$  and  $(P_B, W_B)$ , we could slightly raise  $P_B$  and all the constraints would remain satisfied, since recall that (PC B) is redundant, and the airline would increase its profit coming from business types. Therefore, in an optimal contract, business types are indifferent between  $(P_T, W_T)$  and  $(P_B, W_B)$ : this is another way of expressing that (IC B) is *binding*,  $\theta_B P_B + W_B = \theta_B P_T + W_T$ .

- (d) From part (b) and (c), consider the binding constraints:

$$v = \theta_T P_T + W_T \tag{32}$$

$$\theta_B P_B + W_B = \theta_B P_T + W_T \tag{33}$$

These two constraints together with monotonicity ( $P_B \geq P_T$ ) and single crossing, imply that

$$W_T - W_B = \theta_B P_B - \theta_B P_T = \theta_B (P_B - P_T) \leq \theta_T (P_B - P_T)$$

which is the constraint (PC T).

Thus, once again, we can solve the airline's problem by considering just the binding constraints, the monotonicity constraint, and the non-negativity constraints. From the binding constraints, we will solve for the prices in terms of the remaining waiting time:

$$P_T = \frac{v - W_T}{\theta_T} \quad (34)$$

$$P_B = \frac{v - W_T}{\theta_T} + \frac{W_T}{\theta_B} \quad (35)$$

Note that since  $P_T \geq 0$  and  $P_B \geq 0$  imply from the previous equations that  $W_T \leq v$ . Plugging in the previous expressions into the objective of the airline yields the following problem:

$$\max_{W_T} \left\{ \lambda \left( \frac{v - W_T}{\theta_T} \right) + (1 - \lambda) \left( \frac{v - W_T}{\theta_T} + \frac{W_T}{\theta_B} \right) - c \right\} \quad (36)$$

$$\text{subject to } 0 \leq W_T \leq v \quad (37)$$

Note that monotonicity of the prices is automatically satisfied from equations (34) and (35).

We can simplify the objective (36) to get:

$$\frac{v - W_T}{\theta_T} + (1 - \lambda) \frac{W_T}{\theta_B} = \left( \frac{v}{\theta_T} \right) - c = \left( \frac{v}{\theta_T} - c \right) + W_T \left( \frac{1 - \lambda}{\theta_B} - \frac{1}{\theta_T} \right)$$

Remember that we will eventually need to check if it is worth for the firm to operate, but for the moment, the problem of a firm that decides to operate can be reduced to:

$$\max_{W_T \in [0, v]} \left\{ \left( \frac{v}{\theta_T} - c \right) + W_T \left( \frac{1 - \lambda}{\theta_B} - \frac{1}{\theta_T} \right) \right\} \quad (38)$$

If we analyze problem (38) from a purely mathematical perspective, we can notice that the objective function is an *affine function*, i.e. a line with an intercept, and that the maximization is carried out over an interval. Thus, there are only two possible solutions, which depend on the *sign* of the slope  $\left( \frac{1 - \lambda}{\theta_B} - \frac{1}{\theta_T} \right)$ . The solution is given by:

$$W_T = \begin{cases} v & \text{if } \frac{\theta_T - \theta_B}{\theta_T} \geq \lambda \\ 0 & \text{if } \frac{\theta_T - \theta_B}{\theta_T} < \lambda \end{cases}$$

Before solving for the remaining variables, we need to check under what circumstances the previous is indeed a *true solution* by checking if the firm is making profits. Thus, we will compute first the profits under the two cases above.

$$\text{Profits when } W_T = 0 : \quad \frac{v}{\theta_T} - c$$

$$\text{Profits when } W_T = v : \quad \frac{(1 - \lambda)v}{\theta_B} - c$$

The firm decides to operate whenever the parameters are such that:

$$\min \left\{ \frac{v}{\theta_T}, (1 - \lambda) \frac{v}{\theta_B} \right\} \geq c$$

Whenever the above condition is satisfied, then the remaining variables are given by

$$P_T^* = \begin{cases} 0 & \text{if } \frac{\theta_T - \theta_B}{\theta_T} \geq \lambda \\ \frac{v}{\theta_T} & \text{if } \frac{\theta_T - \theta_B}{\theta_T} < \lambda \end{cases}$$

$$P_B^* = \begin{cases} \frac{v}{\theta_B} & \text{if } \frac{\theta_T - \theta_B}{\theta_T} \geq \lambda \\ \frac{v}{\theta_T} & \text{if } \frac{\theta_T - \theta_B}{\theta_T} < \lambda \end{cases}$$

If the firm decides not to operate, then all the simplifications above (items b–d) do not apply and the optimal contract is  $\{(P_T^*, W_T^*), (P_B^*, W_B^*)\} = \{(0, 0), (0, 0)\}$

Summarizing all these results, the optimal contract is:

$$\{(P_T^*, W_T^*), (P_B^*, W_B^*)\} = \begin{cases} \{(0, v), (\frac{v}{\theta_B}, 0)\} & \text{if } \min\left\{\frac{\theta_T - \theta_B}{\theta_T}, \frac{v - c\theta_B}{v}\right\} \geq \lambda \\ \{(\frac{v}{\theta_T}, 0), (\frac{v}{\theta_T}, 0)\} & \text{if } \lambda > \frac{\theta_T - \theta_B}{\theta_T} \text{ and } \frac{v}{\theta_T} \geq c \\ \{(0, 0), (0, 0)\} & \text{if } \min\left\{\frac{v}{\theta_T}, (1 - \lambda)\frac{v}{\theta_B}\right\} < c \end{cases} \quad (39)$$

(e) From the optimal solution in (39) we can see that whenever

$$\min\left\{\frac{\theta_T - \theta_B}{\theta_T}, \frac{v - c\theta_B}{v}\right\} \geq \lambda$$

, the airline is only contracting with the business type:

$$\{(P_T^*, W_T^*), (P_B^*, W_B^*)\} = \left\{(0, v), \left(\frac{v}{\theta_B}, 0\right)\right\}$$

since the price it charges for tourists is zero. If this is the case, we are in a *shutdown contract*.

On the other hand, whenever

$$\lambda > \frac{\theta_T - \theta_B}{\theta_T} \text{ and } \frac{v}{\theta_T} \geq c$$

the airline optimal plan is a *pooling contract*:

$$\{(P_T^*, W_T^*), (P_B^*, W_B^*)\} = \left\{\left(\frac{v}{\theta_T}, 0\right), \left(\frac{v}{\theta_T}, 0\right)\right\}.$$

In the case that the firm can never make positive profits, i.e. when

$$\min\left\{\frac{v}{\theta_T}, (1 - \lambda)\frac{v}{\theta_B}\right\} < c,$$

then the optimal contract is also a pooling contract in which no traveler gets serviced.

5. Let the agent be a drinker who plans to buy at most one bottle of wine within a month. His utility is  $U = \theta q - t$  where  $q$  is the quality he buys and  $\theta$  is a positive parameter that indexes his taste for quality. If he decides not to buy any wine, his utility is just 0. Suppose that there are  $n$  types of drinkers and that  $\theta_1 < \dots < \theta_n$ . Their respective prior probabilities are  $\pi_1, \dots, \pi_n$  with  $\sum_{i=1}^n \pi_i = 1$ . The principal is a local monopoly on the wine market. He can produce wine with any quality  $q \geq 0$ ; the production of a bottle of wine with quality  $q$  costs him  $c(q) = \frac{q^2}{2}$ .

(a) Show that the only binding constraints are the downward adjacent incentive constraints, i.e.

$$\theta_i q_i - t_i \geq \theta_i q_{i-1} - t_{i-1} \quad i = 2, \dots, n$$

and the participation constraint for the lowest type, i.e.

$$\theta_1 q_1 - t_1 \geq 0$$

(b) Characterize as far as you can the solution to the monopolist's problem.

For the remainder of the problem, assume that  $n = 3$  and that  $\theta_1 = 1, \theta_2 = 5, \theta_3 = 9$ .

- (c) Suppose that  $\pi_1 = 0.85, \pi_2 = 0.08$  and that  $\pi_3 = 0.07$ . Compute the optimal qualities for each consumer, the optimal transfers and the monopolist's profits.
- (d) Suppose now that  $\pi_1 = \pi_2 = \pi_3 = 1/3$ . Compute again the optimal qualities, transfers and the monopolist's profits.
- (e) Compare the results from (c) and (d). What did you find? In which of the two cases is the monopolist performing better? Can you provide an economic interpretation of your results?

### Solution

(a) We start with an approach similar to the one in the lecture slides. The monopolist's problem is

$$\max_{\{(q_i, t_i)\}_{i=1, \dots, n}} \sum_{i=1}^n \pi_i \left( t_i - \frac{q_i^2}{2} \right) \quad (40)$$

subject to

$$\theta_i q_i - t_i \geq \theta_i q_j - t_j, \forall i, j \quad (41)$$

$$\theta_i q_i - t_i \geq 0, \forall i \quad (42)$$

$$q_i \geq 0, \forall i \quad (43)$$

The previous problem is equivalent to the relaxed problem:

$$\max_{\{(q_i, t_i)\}_{i=1, \dots, n}} \sum_{i=1}^n \pi_i \left( t_i - \frac{q_i^2}{2} \right) \quad (44)$$

subject to

$$\theta_i q_i - t_i \geq \theta_i q_{i-1} - t_{i-1}, \forall i = 2, \dots, n \quad (45)$$

$$\theta_1 q_1 - t_1 \geq 0, \quad (46)$$

$$q_i \geq q_{i-1}, \forall i = 2, \dots, n \quad (47)$$

$$q_i \geq 0, \forall i \quad (48)$$

First we will show that when monotonicity (47) holds, then all the incentive constraints (41) hold if and only if the downward constraints (45) and the upward constraints

$$\theta_i q_i - t_i \geq \theta_i q_{i+1} - t_{i+1}, \forall i = 1, \dots, n-1, \quad (49)$$

hold. Clearly the full set of incentive constraints (41) imply both the downward (45) and upward (49) constraints. To prove the other direction, fix an index  $i \in \{2, \dots, n\}$ . Now we need to pick an arbitrary index  $j$  but as mentioned in class, we have two cases:

- i.  $j < i$ : This case was done in class and I am going to skip it. Follow the lecture notes to prove this case.
- ii.  $j > i$ : This is the case that I asked you to verify in the lecture notes. I am going to prove this case here.

Thus, take any  $j > i$ . Notice that

$$\theta_i q_i - q_j \theta_i = \sum_{k=1}^{j-1} \theta_i q_k - \theta_i q_{k+1} \geq \sum_{k=1}^{j-1} \theta_k q_k - \theta_k q_{k+1} \geq \sum_{k=1}^{j-1} t_k - t_{k+1} = t_i - t_j,$$

where the first equality is by a telescopic sum argument, the second inequality is by increasing differences, the third inequality is by upward incentive compatibility (49) and the fourth equality is again by a telescopic sum argument.

We continue with the proof. We proceed in 4 steps:

**Step 1** The constraint set of P is a subset of RP. Notice that the full set of incentive compatibility restrictions (41) imply that

$$\theta_{i-1} q_i - \theta_{i-1} q_{i-1} \leq t_i - t_{i-1} \leq \theta_i q_i - \theta_i q_{i-1}$$

which implies  $(q_i - q_{i-1})(\theta_{i-1} - \theta_i) \geq 0$ . Since types are such that  $\theta_i \geq \theta_{i-1}$  (this is the SCP in for this model), then we have that the previous inequality implies that  $q_i \geq q_{i-1}$ .

**Step 2** At the solution to RP, (45) holds as an equality for all  $i$  and (46) also holds as an equality. Otherwise, we can increase all  $t_j$ ,  $j \geq i$  or  $t_1$  by some  $\epsilon > 0$ , increasing the profit of the monopolist.

**Step 3** Solutions to RP satisfy the constraints of P. Since (45) hold with equality and  $q_i \geq q_{i-1}$ , we have that

$$t_i - t_{i-1} = \theta_i q_i - \theta_i q_{i-1} \geq \theta_{i-1} q_i - \theta_{i-1} q_{i-1},$$

that is, all upward IC constraints (49) hold, and by the first result we proved here, this implies that the full set of IC constraints (41) hold.

**Step 4** The participation constraints of the lowest type (46) implies all participation constraints (42). Note that the full set of IC (41), PC for the lowest type (46), and single crossing ( $\theta_i \geq \theta_{i-1}$ ) imply that

$$\theta_i q_i - t_i \geq \theta_i q_1 - t_1 \geq \theta_1 q_1 - t_1 = 0.$$

Thus we have shown that at an optimal solution, the only relevant constraints are (45) and (46) and that these need to hold with equality, thus completing the proof.

(b) We can follow the general development in the slides to characterize the solution. Foregoing the monotonicity constraint in the relaxed problem, we get:

$$q_n = \theta_n, \quad (50)$$

$$q_i = \max \left\{ \theta_i - \frac{1 - \Pi_i}{\pi_i} (\theta_{i+1} - \theta_i) \right\}, \quad (51)$$

where  $\Pi_i = \sum_{k=1}^i \pi_k$ . We can see from (50) that there is no distortion for the top type, and from (51) there will be a distortion. Furthermore, the expression in (51) takes into consideration the possibility of *corner solutions*, i.e. when the first order condition identifies a negative solution for  $i$ , then we put  $q_i = 0$ .

Note that in order for the solution to be monotonic, we would have to impose more restrictions on the parameters. Let  $\Delta_i = \theta_{i+1} - \theta_i$  for every  $i = 1, \dots, n-1$ , and let  $H_i = \frac{1 - \Pi_i}{\pi_i}$  for  $i = 1, \dots, n$

- A *necessary and sufficient* condition for monotonicity is:  $\frac{\Delta_i}{\Delta_{i+1}} \geq \frac{H_{i+1}}{1 + H_i}$  for  $i = 1, \dots, n-1$ .
- *Sufficient conditions* for monotonicity are that  $H_i$  should be decreasing with  $i$  and that also  $\Delta_i$  should be decreasing with  $i$ . These two conditions are the ones that I mentioned in the last slide of the lecture: the inverse hazard rate should be decreasing in  $i$  (monotone hazard rate property) and that the difference in marginal values between types adjacent types should also be decreasing in  $i$ .

To characterize the transfers, we use the binding constraints (46) and (45):

$$t_1 = \theta_1 q_1, \quad (52)$$

$$t_i = \theta_i (q_i - q_{i-1}) + t_{i-1}, \quad i = 2, \dots, n. \quad (53)$$

(c) Quantities, transfers and profits are:

$$(q_1, q_2, q_3) = \left( \frac{5}{17}, \frac{3}{2}, 9 \right), \quad (54)$$

$$(t_1, t_2, t_3) = \left( \frac{5}{17}, \frac{215}{34}, \frac{1255}{17} \right), \quad (55)$$

$$\text{Profits} = \frac{1007}{340}. \quad (56)$$

(d) Quantities, transfers and profits are:

$$(q_1, q_2, q_3) = (0, 1, 9), \quad (57)$$

$$(t_1, t_2, t_3) = (0, 5, 77), \quad (58)$$

$$\text{Profits} = \frac{41}{3}. \quad (59)$$

(e) The monopolist performs better under (d) than in (c). Notice that the monopolist is shutting down consumption for the lowest type under scheme (d). Thus, at the cost of not contracting with the lowest type, the monopolist is able to extract more rent from higher types, and this trade-off is optimal.

Define the *cumulative distribution functions* for each case as

$$G(\theta) = \begin{cases} 0 & \theta < \theta_1 \\ 0.85 & \theta_1 \leq \theta < \theta_2 \\ 0.93 & \theta_2 \leq \theta < \theta_3 \\ 1 & \theta_3 \leq \theta \end{cases} \quad F(\theta) = \begin{cases} 0 & \theta < \theta_1 \\ 1/3 & \theta_1 \leq \theta < \theta_2 \\ 2/3 & \theta_2 \leq \theta < \theta_3 \\ 1 & \theta_3 \leq \theta \end{cases}$$

Note that  $F(\theta) \leq G(\theta)$  for every  $\theta$ . Intuitively, this means that  $F$  assigns higher probabilities to higher types than  $G$ . This means that there is a higher chance of meeting a higher type than a lower type from  $F$  than from  $G$ . Because of this, the monopolist is confident that by shutting down contracting with the lowest type under  $F$  is profitable.

6. Let the agent be a drinker who plans to buy at most one bottle of wine within a month. His utility is  $U = \theta q - t$  where  $q$  is the quality he buys and  $\theta$  is a positive parameter that indexes his taste for quality. If he decides not to buy any wine, his utility is just 0. This time, suppose that the taste for quality  $\theta$  is uniformly distributed on the interval  $[\theta_L, \theta_H]$ , where  $0 < \theta_L < \theta_H$

The principal is a local monopoly on the wine market. He can produce wine with any quality  $q \geq 0$ ; the production of a bottle of wine with quality  $q$  costs him  $c(q) = \frac{q^2}{2}$ .

- (a) Characterize the second-best optimal quality-pricing schedule.
- (b) Now suppose that  $\theta$  is distributed on  $[\theta_L, \theta_H]$  according to the distribution  $f(\theta) = \frac{2(\theta_H - \theta)}{(\theta_H - \theta_L)^2}$ . Characterize the second-best optimal quality-pricing schedule in this case. Is the hazard rate function  $\frac{1-F(\theta)}{f(\theta)}$  decreasing? What does a decreasing hazard rate has to do with the monotonicity of the quality-pricing schedule?

### Solution

This is a monopoly screening problem. We covered the general case during my lectures so I will skip several details and refer you to my slides when needed. I will solve the problem first for an arbitrary distribution  $f(\theta)$ . The monopolist problem is

$$\begin{aligned} \max_{(q,t)} \quad & \int_{\theta_L}^{\theta_H} \left( t(\theta) - \frac{q(\theta)^2}{2} \right) f(\theta) d\theta \\ \text{subject to} \quad & \theta q(\theta) - t(\theta) \geq 0 \quad \forall \theta \in [\theta_L, \theta_H] \\ & \theta q(\theta) - t(\theta) \geq \theta q(\theta') - t(\theta') \quad \forall \theta, \theta' \in [\theta_L, \theta_H] \end{aligned}$$

For any contract  $(q, t)$  define  $U(\theta) = \theta q(\theta) - t(\theta)$ , i.e., the utility from the consumer of choosing a contract designed for his true type. A contract  $(q, t)$  is incentive compatible if and only if

- (a)  $q$  is monotone increasing
- (b)  $U(\theta) = U(\theta') + \int_{\theta'}^{\theta} q(\tau) d\tau$ .

You can check my slides for the proof of why the two conditions above are equivalent to incentive compatibility. Note that the above conditions imply that the only participation constraint that matters is the one for the lowest type  $\theta_L$ . Once again, for the proof you can refer to the slides.

The problem of the monopolist is equivalent to

$$\begin{aligned} \max_{(q,t)} \quad & \int_{\theta_L}^{\theta_H} \left( t(\theta) - \frac{q(\theta)^2}{2} \right) f(\theta) d\theta \\ \text{subject to} \quad & U(\theta) = U(\theta_L) + \int_{\theta_L}^{\theta} q(\tau) d\tau \\ & q \text{ is increasing} \\ & U(\theta_L) \geq 0. \end{aligned}$$

Note that we can write using the envelope condition the transfers as  $t(\theta) = \theta q(\theta) - U(\theta_L) - \int_{\theta_L}^{\theta} q(\tau) d\tau$ . We can use this fact to rewrite the objective function, which after a lengthy simplification (in the slides, it is the part that involves simplifying the double integral that appears) yields the following problem for the principal

$$\begin{aligned} \max_q \quad & \int_{\theta_L}^{\theta_H} \left( \theta q(\theta) - \frac{q(\theta)^2}{2} - \frac{1 - F(\theta)}{f(\theta)} q(\theta) \right) f(\theta) d\theta \\ \text{subject to} \quad & q \text{ increasing} \end{aligned}$$

As usual, we will solve the problem by first ignoring the monotonicity condition and then identifying conditions under which it will hold.

We can maximize the above integral by choosing  $q$  such that it maximizes *pointwise* the integrand, which yields the first order condition:

$$\theta - q(\theta) - \frac{1 - F(\theta)}{f(\theta)} = 0.$$

Solving for  $q$  yields

$$q(\theta) = \max \left\{ \theta - \frac{1 - F(\theta)}{f(\theta)}, 0 \right\},$$

where I the max takes care of the possibility of negative solutions. Define  $\theta^*$  as the value that solves the following equation:

$$\theta^* = \frac{1 - F(\theta^*)}{f(\theta^*)}.$$

Then we can express  $q(\theta)$  as

$$q(\theta) = \begin{cases} \theta - \frac{1 - F(\theta)}{f(\theta)} & \theta \in [\theta^*, \theta_H] \\ 0 & \theta \in [\theta_L, \theta^*] \end{cases}$$

Thus, notice that the contract exhibits shutdown for some types.

To calculate the transfers we first compute  $U(\theta)$  :

$$U(\theta) = \int_{\theta_L}^{\theta} \max \left\{ \tau - \frac{1 - F(\tau)}{f(\tau)}, 0 \right\} d\tau = \begin{cases} 0 & \theta \in [\theta_L, \theta^*] \\ \int_{\theta^*}^{\theta} \left( \tau - \frac{1 - F(\tau)}{f(\tau)} \right) d\tau & \theta \in [\theta^*, \theta_H] \end{cases}$$

Thus, transfers are  $t(\theta) = \theta q(\theta) - U(\theta)$  where  $q$  and  $U$  are given by the formulas above.

(a) For the uniform distribution we have that

$$\begin{aligned} f(\theta) &= \frac{1}{\theta_H - \theta_L} \\ F(\theta) &= \frac{\theta - \theta_L}{\theta_H - \theta_L} \\ \frac{1 - F(\theta)}{f(\theta)} &= \theta_H - \theta. \end{aligned}$$

Thus  $\theta^* = \frac{\theta_H}{2}$  so that

$$\begin{aligned} q(\theta) &= \begin{cases} 0 & \theta \in [0, \theta_H/2] \\ 2\theta - \theta_H & \theta \in [\theta_H/2, \theta_H] \end{cases} \\ t(\theta) &= \begin{cases} 0 & \theta \in [0, \theta_H/2] \\ \theta^2 - \frac{\theta_H^2}{4} & \theta \in [\theta_H/2, \theta_H] \end{cases} \end{aligned}$$

(b) For this case we have

$$f(\theta) = \frac{2(\theta_H - \theta)}{(\theta_H - \theta_L)^2}$$

$$F(\theta) = \frac{2\theta_H\theta - \theta^2 - 2\theta_H\theta_L + \theta_L^2}{(\theta_H - \theta_L)^2}$$

$$\frac{1 - F(\theta)}{f(\theta)} = \frac{\theta_H - \theta}{2}.$$

Thus  $\theta^* = \frac{\theta_H}{3}$  so that

$$q(\theta) = \begin{cases} 0 & \theta \in [0, \theta_H/3] \\ \frac{3\theta - \theta_H}{2} & \theta \in [\theta_H/3, \theta_H] \end{cases}$$

$$t(\theta) = \begin{cases} 0 & \theta \in [0, \theta_H/3] \\ \frac{9\theta^2 - \theta_H^2}{12} & \theta \in [\theta_H/3, \theta_H] \end{cases}$$

Notice that the hazard rate for this case is decreasing. Recall that a decreasing hazard rate is part of the sufficient conditions required for monotonicity of  $q$ . For the case in this item,  $q$  is indeed increasing in the type.

7. A government agency writes a procurement contract with a firm to deliver  $q$  units of a good. The firm has constant marginal cost  $c$ , so that its profit is  $P - cq$ , where  $P$  denotes the payment for the transaction. The firm's cost is private information and the agency's prior belief about it is uniformly distributed on the interval  $[\underline{c}, \bar{c}]$ . The agency makes a take-it-or-leave-it offer to the firm (whose default profit is zero). Suppose that  $B(q)$  denotes the benefit to the agency of obtaining  $q$  units.  $B(q)$  is a strictly increasing and a strictly concave function of  $q$ .

- (a) Characterize the first-best contract.  
(b) Characterize the second-best contract.

### Solution

- (a) Let  $(q(c), P(c))$  be a contract. At the first-best, the principal solves, for any  $c \in [\underline{c}, \bar{c}]$

$$\begin{aligned} \max_{q(c), P(c)} \quad & B(q(c)) - P(c) \\ \text{subject to} \quad & P(c) - cq(c) \geq 0. \end{aligned}$$

At an optimal solution, the participation constraint must be binding, i.e.,  $P(c) = cq(c)$ . Substituting the binding constraint into the objective yields, for any  $c \in [\underline{c}, \bar{c}]$ , the following problem

$$\max_{q(c)} B(q(c)) - cq(c)$$

The first order condition for the above problem is

$$B'(q(c)) = c \quad \forall c \in [\underline{c}, \bar{c}]$$

which characterizes the first-best solution for the principal.

- (b) I will use here a general distribution function  $f(c)$  and will specialize to the case of the uniform later. The optimal second-best contract is given by the solution to

$$\begin{aligned} \max_{q(c), P(c)} \quad & \int_{\underline{c}}^{\bar{c}} (B(q(c)) - P(c)) f(c) dc \\ & P(c) - cq(c) \geq 0, \quad \forall c \\ & P(c) - cq(c) \geq P(c') - cq(c'), \quad \forall c, c' \end{aligned}$$

Define  $U(c) = P(c) - cq(c)$ . I will first show that incentive compatibility constraints are satisfied if and only if

- i.  $q$  is decreasing
- ii.  $U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau$  for any  $c$

*Proof.* Using the definition of  $U(c)$  we can write the incentive compatibility constraints for any  $c' > c$  as follows:

$$\begin{aligned} U(c) &\geq U(c') + q(c')(c' - c) \\ U(c') &\geq U(c) + q(c)(c - c') \end{aligned}$$

Equivalently,

$$q(c')(c' - c) \leq U(c) - U(c') \leq q(c)(c' - c)$$

Therefore,  $c' > c$  implies that  $q(c) \geq q(c')$ . Dividing by  $c' - c$  and taking the limit as  $c' \rightarrow c$ , we get

$$\frac{dU(c)}{dc} = -q(c)$$

Therefore

$$U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau$$

Suppose now that  $q$  is decreasing and  $U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau$ . Let  $c' > c$ . we have

$$\begin{aligned} U(c) - U(c') &= \int_{\underline{c}}^{c'} q(\tau) d\tau - \int_{\underline{c}}^c q(\tau) d\tau \\ &= \int_c^{c'} q(\tau) d\tau \\ &\geq q(c')(c' - c) \end{aligned}$$

and hence the IC constraints are satisfied. □

Notice that the participation constraint of type  $\bar{c}$ , i.e.,  $U(\bar{c}) \geq 0$  implies that

$$U(\underline{c}) - \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau = U(\bar{c}) \geq 0.$$

Therefore,

$$U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau \geq \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau - \int_{\underline{c}}^c q(\tau) d\tau = \int_c^{\bar{c}} q(\tau) d\tau \geq 0,$$

Where the first inequality is by the participation constraint of type  $\bar{c}$  and the second inequality is by the fact that we are integrating a positive function. Thus, in other words, the participation constraint of type  $\bar{c}$  implies all the other participation constraints.

Therefore, the problem is equivalent to

$$\begin{aligned} &\max_{q(c), P(c)} \int_{\underline{c}}^{\bar{c}} (B(q(c)) - P(c)) f(c) dc \\ &\text{subject to } U(\bar{c}) \geq 0 \\ &U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau \\ &q \text{ is decreasing} \end{aligned}$$

Using the definition of  $U(c)$  and substituting we can write the objective function as

$$\int_{\underline{c}}^{\bar{c}} \left[ B(q(c)) - cq(c) - U(\underline{c}) + \int_{\underline{c}}^c q(\tau) d\tau \right] f(c) dc$$

Substituting for  $U(\underline{c})$  using

$$U(\underline{c}) = U(\bar{c}) + \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau,$$

we can write the objective function as

$$\begin{aligned} & \int_{\underline{c}}^{\bar{c}} \left[ B(q(c)) - cq(c) - U(\bar{c}) - \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau + \int_{\underline{c}}^c q(\tau) d\tau \right] f(c) dc \\ &= \int_{\underline{c}}^{\bar{c}} \left[ B(q(c)) - cq(c) - \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau \right] f(c) dc - U(\bar{c}) \end{aligned}$$

At a solution, we must have that  $U(\bar{c}) = 0$ . We need now to simplify the double integral

$$\int_{\underline{c}}^{\bar{c}} \left( \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau \right) f(c) dc.$$

Using integration by parts this is equal to

$$\int_{\underline{c}}^{\bar{c}} F(c) q(c) dc.$$

The problem is therefore

$$\int_{\underline{c}}^{\bar{c}} \left[ B(q(c)) - cq(c) - \frac{F(c)}{f(c)} q(c) \right] f(c) dc$$

subject to the monotonicity constraint. As usual, we will ignore the monotonicity constraint and then show that the solution satisfies it. The first order condition for each  $c$  is given by

$$B'(q(c)) = c + \frac{F(c)}{f(c)}$$

Notice that at the lowest cost  $\underline{c}$ , the solution is identical to the first-best as was expected. Now, for the uniform distribution,  $F(c) = \frac{c-\underline{c}}{\bar{c}-\underline{c}}$  and  $F(c)/f(c) = c - \underline{c}$  so that

$$B'(q(c)) = c + c - \underline{c} = 2c - \underline{c}$$

Therefore

$$q(c) = (B')^{-1}(2c - \underline{c})$$

Now, if  $B'(q) = b$  then  $((B')^{-1})'(b) = \frac{1}{B''(q)} = \frac{1}{B''((B')^{-1}(b))} < 0$  for any  $b$  since  $B'' < 0$ . Thus, we have that

$$\frac{dq(c)}{dc} = ((B')^{-1})'(2c - \underline{c}) (2) < 0$$

and therefore,  $q$  is strictly decreasing.

Notice also that since  $B' > 0$ , we have that  $q > 0$  for every  $c$ . Finally, note that we can compute the transfers by first calculating  $U(c)$  and then using  $P(c) = U(c) + cq(c)$ .